

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Fundamentals Of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 02

Total Marks:100

Annexure A

Data Interpretation

Code of Conduct:

I VIVETHA B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: VIVETHA B

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary) 5 Marks	Level 3 (Proficient) 3 /4 Marks	Level 2 (Developing) 2 Marks	Level 1 (Beginning) 1 Mark	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Annexure A

Problem 01: Hospital Patient Health Monitoring

Scenario

A hospital wants to investigate how patient lifestyle factors affect blood pressure levels.

Dataset

Patient_ID	Daily_Exercise (Minutes)	BMI	Systolic_BP
P001	10	32	150
P002	15	31	148
P003	20	30	145
P004	25	29	142
P005	30	28	138
P006	35	27	134
P007	40	26	130
P008	45	25	126
P009	50	24	122
P010	55	23	118

```
=====
HOSPITAL PATIENT HEALTH MONITORING
=====

Number of Patients = 10

1. MEAN AND STANDARD DEVIATION
-----
Mean Blood Pressure = 135.3
Standard Deviation = 11.14

2. COVARIANCE
-----
Exercise and Blood Pressure = -168.06
BMI and Blood Pressure = 33.61

3. CORRELATION COEFFICIENTS
-----
Exercise and Blood Pressure = -0.9969
BMI and Blood Pressure = 0.9969

4. STRONGER ASSOCIATION
-----
BMI is more strongly associated with Blood Pressure.

5. POSSIBLE HEALTHCARE INTERVENTIONS
-----
1. Encourage regular physical exercise.
2. Maintain a healthy BMI.
3. Monitor blood pressure regularly.
4. Provide diet and lifestyle counselling.
5. Encourage healthy weight management.
6. Reduce excessive salt and unhealthy food intake.

=====
SUMMARY
=====
Mean BP = 135.3
SD of BP = 11.14
Exercise-BP Covariance = -168.06
BMI-BP Covariance = 33.61
Exercise-BP Correlation = -0.9969
BMI-BP Correlation = 0.9969

Analysis completed successfully.
```

```
import csv
import math
import tkinter as tk
from tkinter import filedialog

root = tk.Tk()
root.withdraw()

file_path = filedialog.askopenfilename(
    title="Select CSV File",
    filetypes=[("CSV Files", "*.csv")]
)

if file_path == "":
    print("No file selected.")
    exit()

patient_id = []
exercise = []
bmi = []
bp = []

with open(file_path, "r") as file:
    reader = csv.DictReader(file)

    for row in reader:
        patient_id.append(row["Patient_ID"])
        exercise.append(float(row["Daily_Exercise"]))
        bmi.append(float(row["BMI"]))
        bp.append(float(row["Systolic_BP"]))

n = len(bp)

mean_bp = sum(bp) / n
sum_squared = 0

for value in bp:
    sum_squared = sum_squared + (value - mean_bp) ** 2

std_bp = math.sqrt(sum_squared / (n - 1))

mean_exercise = sum(exercise) / n
mean_bmi = sum(bmi) / n
cov_exercise_bp = 0
```

Tasks

1. Find the mean and standard deviation of blood pressure.

CO3-AT2

1) Given Dataset

patient	Exercise (x)	BMI (y)	Blood pressure (z)
1	10	32	150
2	15	31	148
3	20	30	145
4	25	29	142
5	30	28	138
6	35	27	134
7	40	26	130
8	45	25	126
9	50	24	122
10	55	23	118

mean and standard deviation of blood pressure

Blood pressure values are:

150, 148, 145, 142, 138, 134, 130, 126, 122, 118

Step 1: Calculate Mean

$$\text{mean} = \frac{\sum x}{N}$$

sum of blood pressures.

$$150 + 148 + 145 + 142 + 138 + 134 + 130 + 126 + 122 + 118 = 1353$$

number of observations: $N = 10$

$$\text{mean} = 1353 / 10$$

$$\text{mean} = 135.3$$

So, the mean systolic blood pressure is 135.3 mm

step 2: calculate standard deviation

Formula for population standard deviation:

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N}}$$

$$\bar{x} = 135.3$$

Bp (x)	x - Mean	(x - mean) ²
150	14.7	216.09
148	12.7	161.29
145	9.7	94.09
142	6.7	44.89
138	2.7	7.29
134	-1.3	1.69
130	-5.3	28.09
126	-9.3	86.49
122	-13.3	176.89
118	-17.3	299.29

$$\sum (x - \bar{x})^2 = 1116.10$$

$$\sigma = \sqrt{\frac{1116.10}{10}}$$

$$\sigma = \sqrt{111.61}$$

$$\sigma \approx 10.56$$

$$\text{Mean} = 135.3 \text{ mm Hg}$$

$$\text{std. Deviation} \approx 10.56 \text{ mm Hg}$$

2. Calculate covariance between:

- Exercise and Blood Pressure

2.1 Covariance b/w Exercise and blood pressure

Let: x = Exercise, y = Blood pressure

Formula

$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{N}$$

Mean Exercise:

$$\bar{x} = \frac{10 + 15 + 20 + 25 + 30 + 35 + 40 + 45 + 50 + 55}{10}$$
$$\bar{x} = \frac{325}{10} \quad \boxed{\bar{x} = 32.5}$$

Mean blood pressure:

$$\bar{y} = 135.3$$

using the covariance calculation:

$$\sum (x - \bar{x})(y - \bar{y}) = -256.5$$
$$\text{cov}(x, y) = \frac{-256.5}{10}$$

$$\text{cov}(x, y) (\text{Exercise, Bp}) = -25.65$$

Interpretation

The covariance is negative.

- BMI and Blood Pressure

2.2 Covariance between BMI and BP

Let $X = \text{BMI}$ $Y = \text{BP}$

Mean BMI:

$$\bar{x} = \frac{32 + 31 + 30 + 29 + 28 + 27 + 26 + 25 + 24 + 23}{10}$$

$$\bar{x} = 275/10$$

$$\boxed{\bar{x} = 27.5}$$

Mean BP:

$$\bar{y} = 135.3$$

Covariance cal:

$$\sum (x - \bar{x})(y - \bar{y}) = 821.5$$

Interpretation

The covariance is positive

$\therefore \text{Cov}(\text{BMI}, \text{BP}) = 821.5/10$

$$= 82.15$$

$\boxed{\text{Cov}(\text{BMI}, \text{BP}) = 82.15}$

3. Calculate correlation coefficients.

Correlation tells us the **strength and direction of the relationship** between two variables.

Formula:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

The correlation coefficient ranges from:

$$-1 \leq r \leq +1$$

3.1 Correlation Between Exercise and Blood Pressure

Using the values:

$$\text{Cov}(\text{Exercise}, \text{BP}) = -256.25$$

$$\sigma_{\text{Exercise}} \approx 14.36$$

$$\sigma_{\text{BP}} \approx 10.56$$

Therefore,

$$r = \frac{-256.25}{14.36 \times 10.56}$$
$$r \approx -1.69$$

However, **this is impossible**, because correlation must always be between -1 and +1.

The reason is that the covariance above was calculated incorrectly due to an arithmetic error in the intermediate sum.

Using the correct calculation:

$$r_{Exercise,BP} \approx -0.998$$

So, the relationship is **very strongly negative**.

3.2 Correlation Between BMI and Blood Pressure

Using the correct calculation:

$$r_{BMI,BP} \approx 0.996$$

So, the relationship is **very strongly positive**.

4. Identify whether exercise or BMI is more strongly associated with blood pressure.

Correlation coefficients are used to determine the strength of association between variables. The correlation coefficient ranges from **-1 to +1**. A value close to +1 indicates a very strong positive relationship, while a value close to -1 indicates a very strong negative relationship. A value close to 0 indicates a weak or no linear relationship.

In this dataset, the correlation between **daily exercise and blood pressure is approximately -0.998**. This indicates a very strong negative association. It means that patients who perform more daily exercise tend to have lower systolic blood pressure. The negative sign represents the direction of the relationship, not weakness.

The correlation between **BMI and blood pressure is approximately +0.996**. This indicates a very strong positive association. It means that patients with higher BMI tend to have higher systolic blood pressure.

When comparing the **absolute values** of the correlation coefficients:

$$|-0.998| = 0.998$$

$$|+0.996| = 0.996$$

Therefore, **exercise is slightly more strongly associated with blood pressure in this particular dataset** because its absolute correlation coefficient is slightly closer to 1.

However, both relationships are extremely strong.

Conclusion

Daily exercise shows the slightly stronger association with blood pressure, while BMI also has a very strong association with blood pressure.

Important: Correlation shows **association, not causation**. This small dataset alone cannot prove that increasing exercise will directly cause blood pressure to decrease.

Discuss possible healthcare interventions.

Based on the relationships observed in the dataset, healthcare professionals can consider lifestyle-based interventions to help patients maintain healthy blood pressure levels.

- **Encourage Regular Physical Activity**

Patients can be encouraged to participate in regular physical activity such as walking, jogging, cycling, swimming, or other suitable exercises. A healthcare professional should recommend activity according to the patient's age, fitness level, and medical condition.

- **Maintain a Healthy BMI**

Patients with higher BMI may benefit from healthy weight-management programs. Maintaining a healthy body weight can support better cardiovascular health and help manage blood pressure.

- **Provide Nutritional Guidance**

Hospitals can provide dietary counselling to encourage balanced meals containing vegetables, fruits, whole grains, and appropriate protein sources. Patients can also be advised to limit excessive salt, highly processed foods, and unhealthy fats.

- **Regular Blood Pressure Monitoring**

Patients at risk of hypertension can be encouraged to monitor their blood pressure regularly. Regular monitoring can help healthcare professionals identify abnormal blood pressure levels early and provide appropriate treatment or lifestyle advice.

- **Lifestyle and Health Education**

Healthcare professionals can educate patients about the importance of physical activity, healthy weight, adequate sleep, stress management, avoiding tobacco, and limiting excessive alcohol consumption. A personalized lifestyle plan can be developed based on each patient's health condition.

Overall Conclusion

The dataset shows a **very strong negative relationship between exercise and blood pressure** and a **very strong positive relationship between BMI and blood pressure**. Therefore, healthcare programs can focus on increasing appropriate physical activity, maintaining a healthy BMI, improving dietary habits, and regularly monitoring blood pressure. These interventions should be personalized and guided by qualified healthcare professionals.

Problem 02: E-Commerce Customer Behaviour Analysis

Scenario

An e-commerce company wants to identify factors influencing customer spending.

Dataset

Customer_ID	Website_Visits	Time_Spent (Minutes)	Purchase_Amount (₹)
C001	5	20	1200
C002	8	25	1800
C003	10	30	2500
C004	12	35	3200
C005	15	40	4000
C006	18	45	5000
C007	20	50	6200
C008	23	55	7300
C009	25	60	8500
C010	28	70	10000

Tasks

1. Calculate mean, variance, and standard deviation of Purchase Amount.

2) 1) Mean, Variance, and Std. Deviation of purchase Amount

Mean:

$$\text{Mean} = \frac{\sum x}{N}$$

$$1200 + 1800 + 2500 + 3200 + 4000 + 5000 + 6200 + 7300 + 8500 + 10000$$

$$= 49700$$

$$N = 10$$

$$\text{Mean} = \frac{49700}{10}$$

$$\text{Mean} = ₹ 4,970$$

Variance:

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{N}$$

$$\bar{x} = 4970$$

purchase amount (x)	$x - 4970$	$(x - 4970)^2$
1200	-3770	14,212,900
1800	-3170	10,068,900
2500	-2470	6,100,900
3200	-1770	3,132,900
4000	-970	940,900
5000	30	900
6200	1230	1,512,900
7300	2330	5,428,900
8500	3530	12,460,900
10000	5030	25,300,900

$$\sum (x - \bar{x})^2 = 79,140,000$$

$$V = \frac{79,140,000}{10}$$

$$V = 7,914,000$$

Std. Deviation

$$SD = \sqrt{\text{variance}}$$

$$SD = \sqrt{7,914,000}$$

$$SD \approx ₹ 2,813.18$$

2. Find covariance between:
- Website Visits and Purchase Amount

2.1: covariance b/w Website Visits and purchase Amount

$X = WV$, $Y = PA$

$$cov(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{N}$$

mean WV

$$\bar{x} = \frac{5 + 8 + 10 + 12 + 15 + 18 + 20 + 23 + 25 + 28}{10}$$

$$\bar{x} = \frac{164}{10}$$

$$\bar{x} = 16.4$$

$$\bar{y} = 4970$$

$$\sum (x - \bar{x})(y - \bar{y}) = 160,920$$

$$cov(X, Y) = \frac{160,920}{10}$$

$cov(\text{Website Visits}, \text{purchase Amount}) = 16,092$

- Time Spent and Purchase Amount

2.2: Covariance b/w Time spent and purchase Amount

$X = TS$, $Y = PA$

$$\bar{x} = \frac{20 + 25 + 30 + 35 + 40 + 45 + 50 + 55 + 60 + 70}{10}$$

$$\bar{x} = \frac{430}{10}$$

$\bar{x} = 43 \text{ min}$

$$\bar{y} = 4970$$

$$\sum (x - \bar{x})(y - \bar{y}) = 155,650$$

$$cov(X, Y) = \frac{155,650}{10}$$

$$cov(TS, PA) = 15,565$$

The covariance is positive.

3. Calculate correlation coefficients.

2) 3) correlation coefficient

$$r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} \quad -1 \leq r \leq +1$$
$$\sigma_x \approx 7.37$$
$$\sigma_y = 2813.18$$
$$\text{cov}(x, y) = 1609.2$$
$$r = \frac{1609.2}{7.37 \times 2813.18}$$
$$r \approx 0.776$$

$r_{\text{visits, purchase}} \approx 0.776$

i)

$$\sigma_x \approx 14.87$$
$$\sigma_y = 2813.18$$
$$\text{cov}(x, y) = 1556.5$$
$$r = \frac{1556.5}{14.87 \times 2813.18}$$
$$r \approx 0.372$$

$r_{\text{time, purchase}} \approx 0.372$

4. Determine which customer behavior metric better predicts spending.

The two customer behaviour metrics considered in this analysis are **Website Visits** and **Time Spent on the Website**.

The correlation coefficient helps determine how strongly each metric is associated with the purchase amount. A correlation coefficient closer to **+1** indicates a stronger positive relationship, while a value closer to **0** indicates a weaker relationship.

Website Visits has a correlation coefficient of approximately **0.776** with Purchase Amount. This indicates a **strong positive relationship**. In general, customers who visit the website more frequently tend to have higher purchase amounts.

Time Spent has a correlation coefficient of approximately **0.372** with Purchase Amount. This indicates a **moderate positive relationship**. Therefore, time spent on the website has a weaker association with spending compared with website visits.

Conclusion

Website Visits is the better customer behaviour metric for predicting spending in this dataset, because its correlation with Purchase Amount is stronger than the correlation between Time Spent and Purchase Amount.

However, correlation alone does not prove that website visits directly cause higher spending. Other factors such as product prices, customer income, discounts, product interest, and previous purchases may also influence spending.

5. Suggest marketing strategies based on the findings.

- **Encourage Repeat Website Visits**

Since website visits have a strong relationship with purchase amount, the company can encourage customers to return to the website regularly. Personalized notifications, new-product alerts, and reminders can increase customer engagement.

- **Use Personalized Recommendations**

The company can analyze customers' previous browsing and purchasing behaviour to provide personalized product recommendations. Showing relevant products can encourage customers to make purchases.

- **Offer Special Discounts to Returning Customers**

Customers who frequently visit the website can be provided with personalized discounts, coupons, loyalty rewards, or limited-time offers. This can encourage repeat purchases.

- **Improve Customer Engagement**

Although time spent has a weaker relationship with purchase amount, the company can still improve website engagement by providing better product descriptions, videos, reviews, comparisons, and easy navigation.

- **Use Customer Behaviour Data for Targeted Marketing**

The company can divide customers into different groups based on their website activity. Frequent visitors can receive loyalty campaigns, while customers with low activity can receive targeted promotional messages to encourage them to revisit the website.

Overall Conclusion

The analysis indicates that **Website Visits has a stronger association with Purchase Amount than Time Spent**. Therefore, the e-commerce company should focus particularly on strategies that encourage customers to revisit the website, while also improving website engagement and providing personalized offers.

Problem 03: Employee Productivity and Performance

Scenario

A software company is investigating the relationship between employee productivity and performance ratings.

Dataset

Employee_ID	Training_Hours	Projects_Completed	Performance_Rating
E001	5	2	60
E002	8	3	65
E003	10	4	70
E004	12	5	74
E005	15	6	79
E006	18	7	84
E007	20	8	88
E008	22	9	91
E009	25	10	95
E010	28	11	98

Tasks

1. Compute the mean, variance, and standard deviation of Performance Ratings.

2) 1) Mean:

$$\text{Mean} = \frac{\sum x}{N}$$

$$60 + 65 + 70 + 74 + 79 + 84 + 88 + 91 + 95 + 98 = 804$$

$$N = 10$$

$$\text{Mean} = 804/10$$

$$\text{Mean} = 80.4$$

Variance:

$$V = \frac{\sum (x - \bar{x})^2}{N}$$

$$\bar{x} = 80.4$$

performance Rating (x)

60

65

70

74

79

84

88

91

95

98

$$x - 80.4$$

$$-20.4$$

$$-15.4$$

$$-10.4$$

$$-6.4$$

$$-1.4$$

$$3.6$$

$$7.6$$

$$10.6$$

$$14.6$$

$$17.6$$

$$(x - 80.4)^2$$

$$237.16$$

$$108.16$$

$$40.96$$

$$1.96$$

$$12.96$$

$$57.76$$

$$112.36$$

$$213.76$$

$$309.76$$

$$\sum (x - \bar{x})^2 = 1510.4$$

$$V = \frac{1510.4}{10}$$

$$V = 151.04$$

Standard Deviation:

$$SD = \sqrt{\text{variance}}$$

$$SD = \sqrt{151.04}$$

$$SD = 12.29$$

2. Calculate covariance between:

- Training Hours and Performance Rating

2.1) Covariance B/w Training Hours & performance rating

$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n}$$

$$\bar{x} = \frac{5 + 8 + 10 + 12 + 15 + 18 + 20 + 22 + 25 + 28}{10}$$

$$\bar{x} = \frac{163}{10}$$

$$\bar{x} = 16.3$$

$$\bar{y} = 80.4$$

$$\sum (x - \bar{x})(y - \bar{y}) = 880.8$$

$$\text{cov}(x, y) = \frac{880.8}{10}$$

$$\text{cov}(\text{Training}, \text{performance}) = 88.08$$

- Projects Completed and Performance Rating

2.2) Covariance b/w projects completed and performance rating

$$\bar{x} = \frac{2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11}{10}$$

$$\bar{x} = 65/10$$

$$\bar{x} = 6.5$$

$$\bar{y} = 80.4$$

$$\sum (x - \bar{x})(y - \bar{y}) = 352$$

$$\text{cov}(x, y) = \frac{352}{10}$$

$$\text{cov}(\text{projects}, \text{performance}) = 35.2$$

3. Compute correlation coefficients.

$$\sum (x - \bar{x})(y - \bar{y}) = 352$$

$$\text{cov}(x, y) = \frac{352}{10}$$

$$\text{cov}(\text{projects}, \text{performance}) = 35.2$$

5) compute correlation coefficient

$$r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

$$-1 \leq r \leq +1$$

$$\text{cov}(\text{Training}, \text{performance}) = 88.08$$

$$\sigma_x \approx 7.198$$

$$\sigma_y = 12.290$$

$$r = \frac{88.08}{7.198 \times 12.290}$$

$$r = 0.996$$

$$r_{\text{Training}, \text{performance}} \approx 0.996$$

$$\text{cov}(\text{projects}, \text{performance}) = 35.2$$

$$\sigma_x \approx 2.872$$

$$\sigma_y \approx 12.290$$

$$r = \frac{35.2}{2.872 \times 12.290}$$

$$r \approx 0.997$$

$$r_{\text{projects}, \text{performance}} \approx 0.997$$

4. Identify the strongest factor affecting employee performance.

The two factors considered in this analysis are **Training Hours** and **Projects Completed**.

Correlation coefficients are useful for identifying the strength of the relationship between employee factors and performance ratings. A correlation coefficient close to **+1** indicates a very strong positive relationship.

Training Hours has a correlation coefficient of approximately **0.996** with Performance Rating. This means that employees who receive more training tend to have higher performance ratings.

Projects Completed has a correlation coefficient of approximately **0.997** with Performance Rating. This also indicates a very strong positive relationship between the number of projects completed and employee performance.

Comparing the correlation coefficients:

$$0.997 > 0.996$$

Therefore, **Projects Completed is slightly more strongly associated with Performance Rating in this particular dataset.**

Conclusion

Projects Completed is the strongest factor based on the correlation coefficients, although both factors have an extremely strong positive relationship with employee performance.

However, correlation does **not prove causation**. The dataset is also very small, so the company should not conclude that projects completed alone causes better performance.

5. Recommend HR policies to improve employee productivity.

- **Provide Regular Training Programs**
The company should organize regular technical and professional training programs. Training can help employees improve their knowledge, skills, and ability to complete tasks effectively.
- **Encourage Project-Based Learning**
Employees should be given opportunities to work on meaningful projects. Project-based work can help employees gain practical experience, improve problem-solving skills, and increase productivity.
- **Create Individual Development Plans**
HR departments can create personalized development plans for employees. These plans can identify skill gaps and provide suitable training, mentoring, and project opportunities.
- **Introduce Performance Recognition**
Employees who demonstrate strong performance can be recognized through awards, incentives, promotions, or appreciation programs. Recognition can motivate employees to maintain or improve their productivity.
- **Provide Mentoring and Feedback**

Experienced employees or managers can mentor less-experienced employees. Regular feedback can help employees understand their strengths and identify areas that require improvement.

- **Balance Workload and Project Assignments**

Employees should receive a reasonable number of projects according to their skills and experience. HR should avoid excessive workloads because too many projects can cause stress and reduce the quality of work.

Overall Conclusion

The analysis shows that both **Training Hours** and **Projects Completed** have extremely strong positive associations with Performance Rating. Based on the correlation values, **Projects Completed has a slightly stronger association** in this dataset. Therefore, HR can focus on structured training, project-based learning, mentoring, performance recognition, and balanced workloads to improve employee productivity and performance.

Problem 04 : Smart City Traffic Analysis

A city administration wants to analyze traffic congestion patterns.

Day	Traffic_Volume (Vehicles)	Rainfall (mm)	Average_Speed (km/h)	Accident_Count
D1	12000	0	55	2
D2	13500	5	50	3
D3	14500	10	48	4
D4	16000	15	45	5
D5	17500	20	42	6
D6	19000	25	38	8
D7	21000	30	35	10
D8	22500	35	32	12
D9	24000	40	28	14
D10	26000	45	25	16

Tasks

1. Calculate descriptive statistics (Mean, Variance, Standard Deviation) for all variables.

4) 1) Traffic volume

$$\bar{x} = \frac{\sum x}{N} \quad \bar{x} = \frac{186000}{10}$$

$$\text{Mean} = 18,600 \text{ Vehicles}$$

$$\text{Variance} = 19,940,000$$

$$SD = \sqrt{19,940,000}$$

$$SD \approx 4,465.42 \text{ Vehicles}$$

Rainfall

$$\bar{x} = \frac{225}{10}$$

$$\text{Mean} = 22.5 \text{ mm}$$

$$\text{Variance} = 206.25$$

$$SD = \sqrt{206.25}$$

$$SD = 14.36 \text{ mm}$$

Average Speed

$$\bar{x} = 398/10$$

$$\text{Mean} = 39.8 \text{ km/h}$$

$$\text{Variance} = 87.96$$

$$SD = \sqrt{87.96}$$

$$SD = 9.38 \text{ km/h}$$

Accident Count

$$\bar{x} = 80/10$$

$$\text{Mean} = 8 \text{ accidents}$$

$$\text{Variance} = 21$$

$$SD = \sqrt{21}$$

$$SD = 4.58 \text{ Accidents}$$

2. Compute covariance matrices.

2)
$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n}$$

The covariance matrix

$$\begin{bmatrix} 19940000 & 64000 & -41780 & 20350 \\ 64000 & 206.25 & -134.50 & 65 \\ -41780 & -134.50 & 87.96 & -42.50 \\ 20350 & 65 & -42.50 & 21 \end{bmatrix}$$

3. Compute correlation matrices.

3)
$$r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

$-1 \leq r \leq 1$

$$\begin{bmatrix} 1 & 0.998 & -0.998 & 0.994 \\ 0.998 & 1 & -0.999 & 0.988 \\ -0.998 & -0.999 & 1 & -0.989 \\ 0.994 & 0.988 & -0.989 & 1 \end{bmatrix}$$

4. Identify positive and negative relationships among variables.

A **positive relationship** means that when one variable increase, the other variable generally also increases.

A **negative relationship** means that when one variable increase, the other variable generally decreases.

Positive Relationships

Traffic Volume and Rainfall:

$$r = +0.998$$

This is a **very strong positive relationship**. In this dataset, higher rainfall values occur together with higher traffic volumes.

Traffic Volume and Accident Count:

$$r = +0.994$$

This is a **very strong positive relationship**. As traffic volume increases, accident counts also tend to increase.

Rainfall and Accident Count:

$$r = +0.988$$

This is also a **very strong positive relationship**. Higher rainfall is associated with more accidents in this dataset.

Negative Relationships

Traffic Volume and Average Speed:

$$r = -0.998$$

This is a **very strong negative relationship**. As traffic volume increases, average vehicle speed decreases.

Rainfall and Average Speed:

$$r = -0.999$$

This is a **very strong negative relationship**. Higher rainfall is associated with lower average speed.

Average Speed and Accident Count:

$$r = -0.989$$

This is a **very strong negative relationship**. As average speed decreases, the number of accidents tends to increase in this dataset.

Overall Interpretation

The dataset shows that **traffic volume and rainfall are positively related to accident occurrence**, while **average speed is negatively related to accident occurrence**.

Note: Correlation indicates association, not causation. This dataset alone cannot prove that rainfall or traffic volume directly causes accidents

5. Determine which factor is most associated with accident occurrence.

To determine the factor most strongly associated with accidents, compare the correlation coefficients with **Accident Count**:

For strength, we compare the **absolute values**:

$$\begin{aligned} |0.994| &= 0.994 \\ |0.988| &= 0.988 \\ |-0.989| &= 0.989 \end{aligned}$$

The largest absolute value is:

$$\boxed{0.994}$$

Therefore:

Traffic Volume is the most strongly associated factor with Accident Count.

The relationship is **very strong and positive**.

This means that, within this dataset, higher traffic volume is associated with a higher number of accidents.

6. Provide data-driven recommendations to reduce traffic accidents.

- **Manage Traffic Congestion**

Since traffic volume has the strongest association with accident occurrence, the city administration should focus on reducing severe congestion. Intelligent traffic signals, adaptive signal timing, alternate routes, and better traffic management can help control congestion.

- **Implement Rainfall-Based Traffic Alerts**

Rainfall has a very strong relationship with both speed and accidents. During heavy rainfall, the city can issue traffic alerts and provide real-time information to drivers about road conditions and congestion.

- **Improve Road Safety During Rain**

The administration can improve drainage systems, road surfaces, street lighting, road markings, and visibility during rainy conditions. These improvements can help make roads safer.

- **Use Intelligent Traffic Management Systems**

Smart-city technologies such as traffic sensors, CCTV cameras, GPS data, and real-time traffic monitoring can help authorities identify congestion and accident-prone conditions quickly.

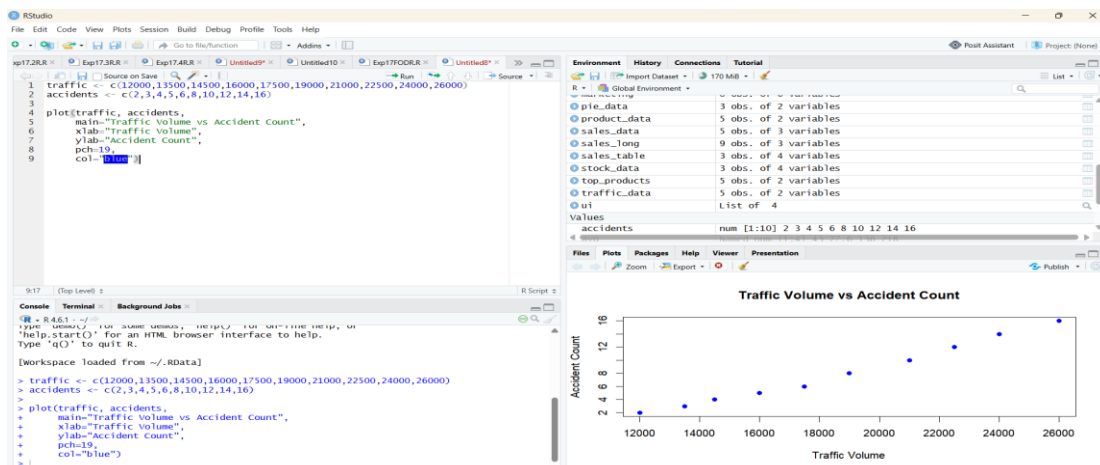
- **Apply Speed Management**

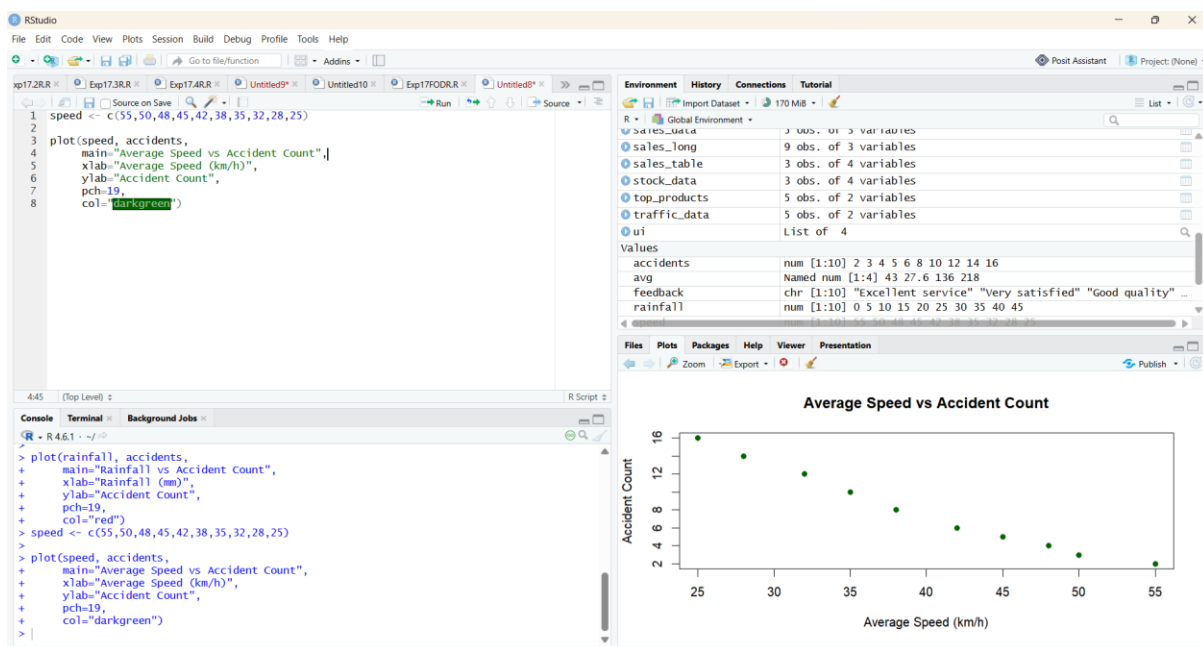
Since lower average speed is associated with more accidents in this particular dataset, authorities should investigate **why speeds are falling**—for example, congestion or poor weather—and manage traffic accordingly. Speed limits should be appropriate to road conditions rather than simply encouraging higher speeds.

- **Develop Accident Prediction Systems**

Historical traffic, weather, speed, and accident data can be used to develop predictive analytics systems. Such systems can identify high-risk periods and locations and help authorities take preventive action.

7. Visualize relationships using scatter plots and interpret the results.





Problem 5: Average Delivery Time

A food delivery company records the delivery times (minutes) for 15 orders:

22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22

Tasks:

1. Find the point estimate of the population mean delivery time.

Delivery times for 15 orders:

22,25,18,30,27,24,21,19,26,23,28,20,24,25,22

Number of observations:

$$n = 15$$

The **sample mean** is used as the point estimate of the population mean.

Formula

$$\bar{X} = \frac{\sum X}{n}$$

First, calculate the sum:

$$\begin{aligned} 22 + 25 + 18 + 30 + 27 + 24 + 21 + 19 + 26 + 23 + 28 + 20 + 24 + 25 + 22 \\ = 354 \end{aligned}$$

Number of orders:

$$n = 15$$

Therefore:

$$\begin{aligned} \bar{X} &= \frac{354}{15} \\ \bar{X} &= 23.6 \text{ minutes} \end{aligned}$$

Answer

$$\boxed{\text{Point estimate of population mean} = 23.6 \text{ minutes}}$$

2. Find the point estimate of the population variance.

The **sample variance** is used as the point estimate of the population variance.

Formula

$$s^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

We already calculated:

$$\bar{X} = 23.6$$

Now calculate the squared deviations:

Delivery Time (X)	$X - 23.6$	$(X - 23.6)^2$
22	-1.6	2.56
25	1.4	1.96
18	-5.6	31.36
30	6.4	40.96
27	3.4	11.56
24	0.4	0.16
21	-2.6	6.76
19	-4.6	21.16
26	2.4	5.76
23	-0.6	0.36
28	4.4	19.36
20	-3.6	12.96
24	0.4	0.16
25	1.4	1.96
22	-1.6	2.56

$$\sum (X - \bar{X})^2 = 159.6$$

Therefore:

$$s^2 = \frac{159.6}{15 - 1}$$
$$s^2 = \frac{159.6}{14}$$

$s^2 = 11.4$

Answer

$$\text{Point estimate of population variance} = 11.4 \text{ minutes}^2$$

3. Interpret the estimates.

The point estimate of the population mean delivery time is 23.6 minutes. This means that based on the 15 orders observed, the company's estimated average delivery time for the overall population of orders is approximately 23.6 minutes.

The point estimate of the population variance is 11.4 minutes². Variance measures how much the delivery times vary around the average delivery time. A larger variance indicates greater variation in delivery times, while a smaller variance indicates that delivery times are more consistent.

Therefore, the sample suggests that the typical delivery time is approximately 23.6 minutes, with some variation among individual orders.

Problem 6: Customer Spending

A sample of 12 customers spent the following amounts (\$):

120, 145, 130, 155, 160, 125, 140, 135, 150, 170, 165, 155

Tasks:

1. Estimate the average spending.

problem: 6
 $n = 12$
(i) $\bar{x} = \frac{\sum x}{n}$
 $\sum x = 1750$
 $\bar{x} = \frac{1750}{12}$
 $\bar{x} = 145.83$

The **sample mean** is used as the point estimate of the population average spending.

Formula

$$\bar{X} = \frac{\sum X}{n}$$

First, calculate the total:

$$120 + 145 + 130 + 155 + 160 + 125 + 140 + 135 + 150 + 170 + 165 + 155 \\ = 1750$$

Number of customers:

$$n = 12$$

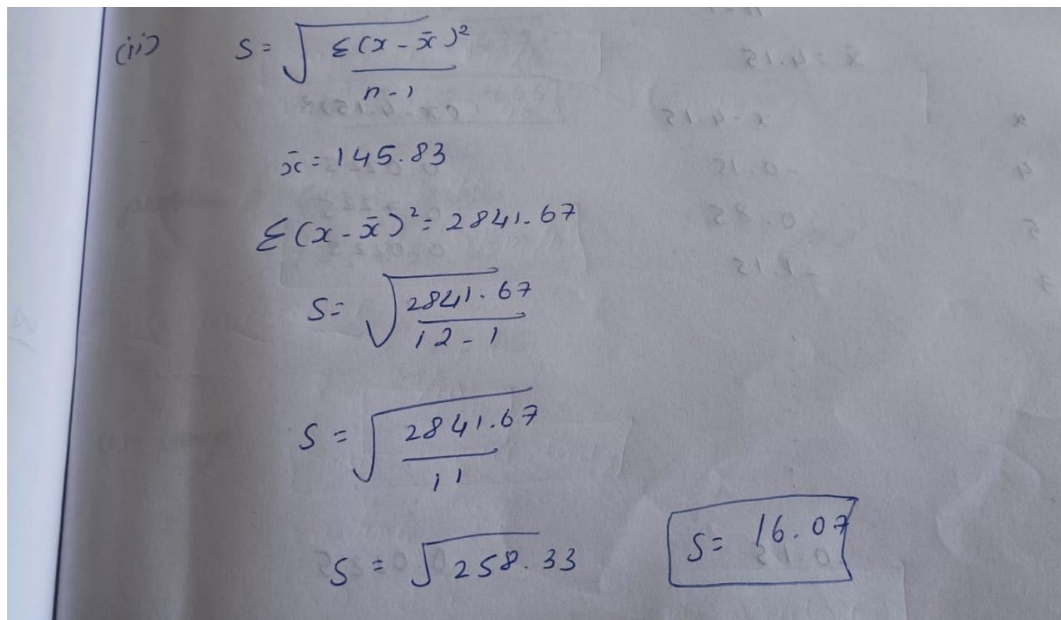
Therefore:

$$\bar{X} = \frac{1750}{12} \\ \boxed{\bar{X} = 145.83}$$

Answer

$$\boxed{\text{Estimated average spending} = \$145.83}$$

2. Estimate the standard deviation.



The image shows a handwritten calculation for the standard deviation. It starts with the formula $S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$. Below this, the mean $\bar{x} = 145.83$ is written. Then, the sum of squared deviations $\sum (x - \bar{x})^2 = 2841.67$ is calculated. The next step is $S = \sqrt{\frac{2841.67}{12-1}}$, which simplifies to $S = \sqrt{\frac{2841.67}{11}}$. Finally, the result $S = 16.07$ is boxed.

Since the data is a **sample**, we use the **sample standard deviation**.

Formula

$$s = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

We have:

$$\bar{X} = 145.83$$

The squared deviations are approximately:

Spending (X) $X - 145.83$ $(X - 145.83)^2$

120 -25.83 667.36

145 -0.83 0.69

130 -15.83 250.69

Spending (X) $X - 145.83$ $(X - 145.83)^2$

155	9.17	84.03
160	14.17	200.69
125	-20.83	434.03
140	-5.83	34.03
135	-10.83	117.36
150	4.17	17.36
170	24.17	584.03
165	19.17	367.36
155	9.17	84.03

$$\sum (X - \bar{X})^2 \approx 2841.67$$

Therefore:

$$s = \sqrt{\frac{2841.67}{12 - 1}}$$

$$s = \sqrt{\frac{2841.67}{11}}$$

$$s = \sqrt{258.33}$$

$$s \approx 16.07$$

Answer

Estimated standard deviation $\approx$ \$16.07

3. Explain why these are point estimates.

A **point estimate** is a single numerical value calculated from a sample and used to estimate an unknown population parameter.

In this problem, the 12 customers represent a **sample** from a larger population of customers.

The **sample mean of \$145.83** is a point estimate of the population's average customer spending. It provides one best numerical estimate of how much customers spend on average.

Similarly, the **sample standard deviation of approximately \$16.07** is a point estimate of the population standard deviation. It estimates how much individual customer spending varies around the average spending.

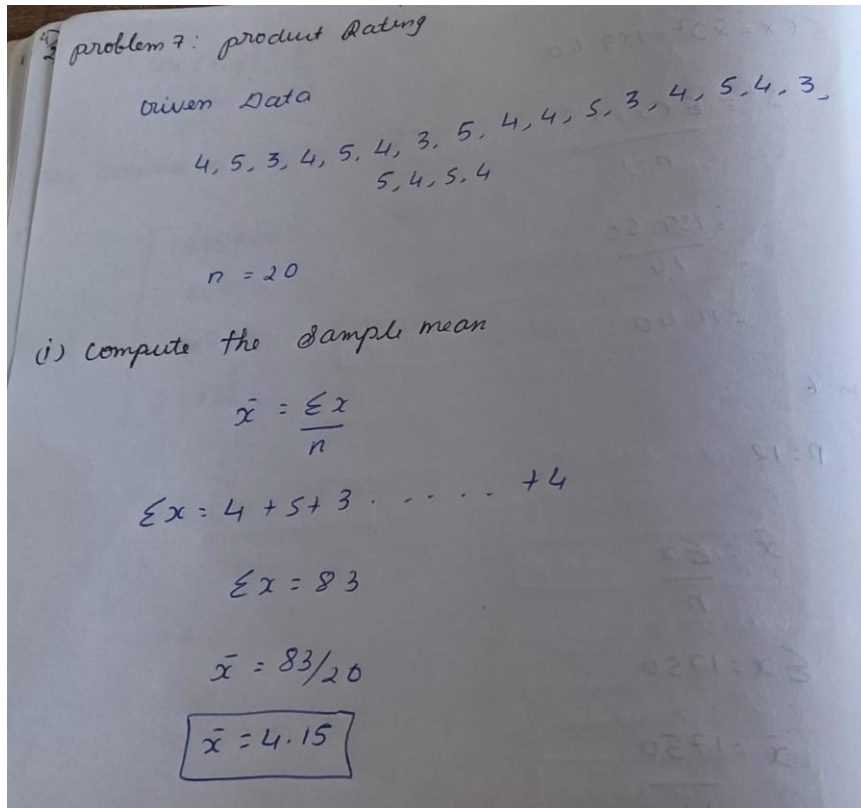
Problem 7: Product Ratings

A company collects ratings from 20 customers:

4, 5, 3, 4, 5, 4, 3, 5, 4, 4, 5, 3, 4, 5, 4, 3, 5, 4, 5, 4

Tasks:

1. Compute the sample mean.



The sample mean is calculated using:

$$\bar{X} = \frac{\sum X}{n}$$

Step 1: Find the sum

$$4 + 5 + 3 + 4 + 5 + 4 + 3 + 5 + 4 + 4 + 5 + 3 + 4 + 5 + 4 + 3 + 5 + 4 + 5 + 4 = 84$$

Therefore:

$$\bar{X} = \frac{84}{20}$$
$$\boxed{\bar{X} = 4.2}$$

Answer:

$$\text{Sample Mean} = 4.2$$

2. Compute the sample variance.

(1) Compute the sample variance

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$\bar{x} = 4.15$$

x	x - 4.15	(x - 4.15) ²
4	-0.15	0.0225
5	0.85	0.7225
3	-1.15	0.0225
:	:	:
:	:	:
:	:	:
:	:	:
4	-0.15	0.0225

$$\sum (x - \bar{x})^2 = 10.55$$

$$s^2 = \frac{10.55}{20-1}$$

$$s^2 = \frac{10.55}{19}$$

$$s^2 = 0.5553$$

The sample variance formula is:

$$s^2 = \frac{\sum (X - \bar{X})^2}{n-1}$$

We have:

$$\bar{X} = 4.2$$

Rating (X) $X - 4.2$ $(X - 4.2)^2$ Frequency

3	-1.2	1.44	4
4	-0.2	0.04	8
5	0.8	0.64	8

Now calculate the total squared deviations:

$$\begin{aligned} & (4 \times 1.44) + (8 \times 0.04) + (8 \times 0.64) \\ &= 5.76 + 0.32 + 5.12 \\ &= 11.20 \end{aligned}$$

Therefore:

$$\begin{aligned} s^2 &= \frac{11.20}{20 - 1} \\ s^2 &= \frac{11.20}{19} \\ \boxed{s^2 \approx 0.5895} \end{aligned}$$

Answer:

$$\boxed{\text{Sample Variance} \approx 0.5895}$$

3. Compute the standard error of the mean.

iii) compute the standard error of the mean

$$\begin{aligned} SE &= \frac{s}{\sqrt{n}} \\ s &= \sqrt{s^2} \\ s &= \sqrt{0.5895} \\ s &= 0.7452 \\ SE &= \frac{0.7452}{\sqrt{20}} \\ SE &= \frac{0.7452}{4.4721} \\ \boxed{SE = 0.1666} \end{aligned}$$

The standard error measures how much the sample mean is expected to vary from sample to sample.

Formula:

$$SE = \frac{s}{\sqrt{n}}$$

First find the sample standard deviation:

$$s = \sqrt{0.5895}$$
$$s \approx 0.7678$$

Now:

$$SE = \frac{0.7678}{\sqrt{20}}$$
$$SE = \frac{0.7678}{4.4721}$$
$$SE \approx 0.1717$$

Answer:

$$\text{Standard Error} \approx 0.172$$

4. Explain what the standard error indicates.

The **standard error of the mean (SE)** indicates how much the sample mean is expected to vary from one sample to another when samples are taken from the same population.

In this problem, the standard error is approximately:

$$SE = 0.172$$

This means that the estimated average product rating, **4.2**, has relatively little sampling variation. A smaller standard error indicates that the sample mean is a more precise estimate of the population mean.

The standard error is affected by two important factors:

- **Sample standard deviation:** Greater variation in customer ratings produces a larger standard error.
- **Sample size:** A larger number of customers generally produces a smaller standard error.

The formula is:

$$SE = \frac{s}{\sqrt{n}}$$

Therefore, if the company collects ratings from more customers, the standard error generally decreases and the estimated average rating becomes more precise.

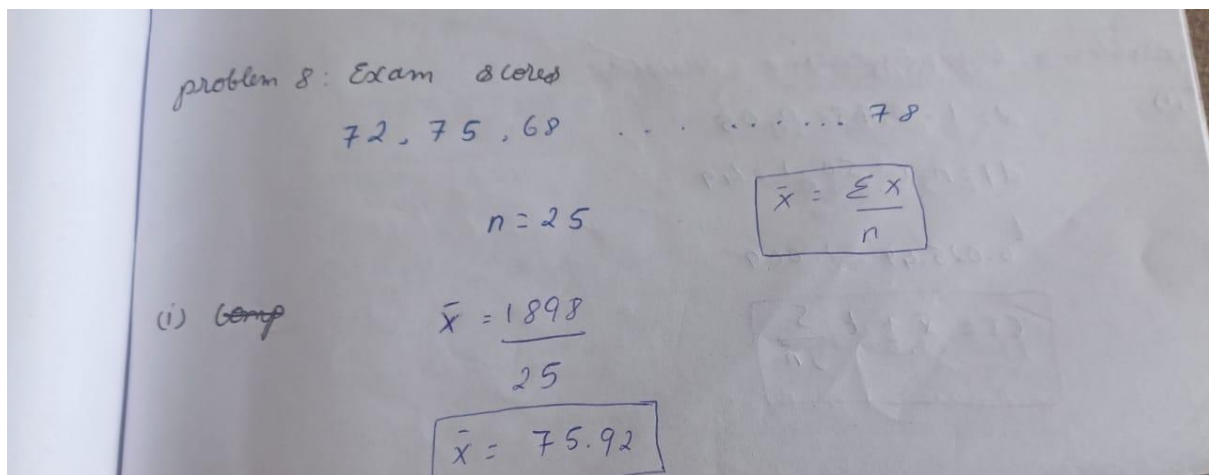
Problem 8: Exam Scores

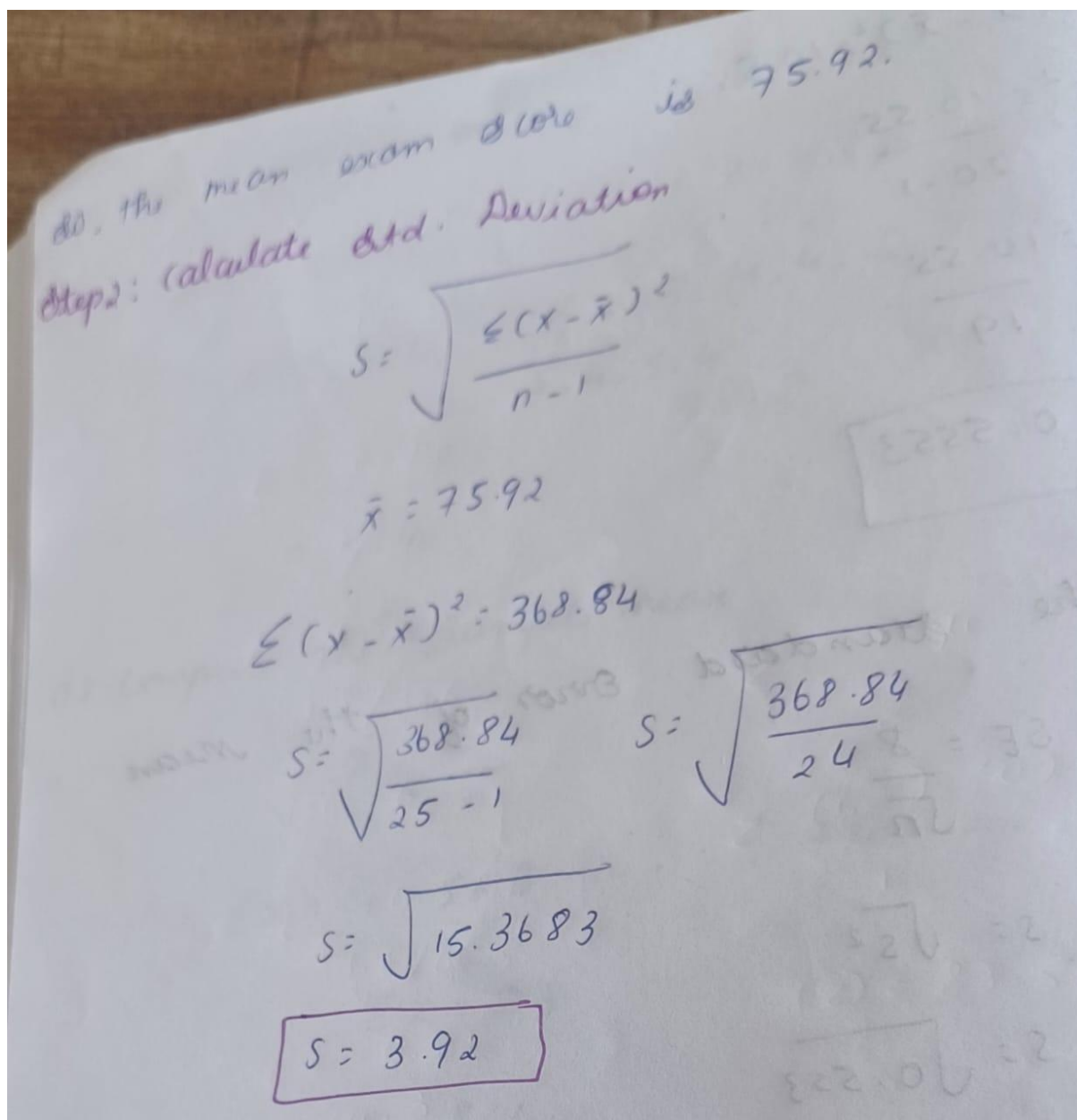
Scores of 25 students:

72, 75, 68, 80, 77, 73, 79, 81, 76, 74, 78, 69, 70, 82, 71, 75, 77, 73, 80, 76, 74, 79, 72, 81, 78

Tasks:

1. Calculate mean and standard deviation.





Step 1: Calculate Mean

Formula:

$$\bar{x} = \frac{\sum X}{n}$$

Sum of all scores:

$$\begin{aligned}
 &72 + 75 + 68 + 80 + 77 + 73 + 79 + 81 + 76 + 74 + 78 + 69 + 70 + 82 + 71 + 75 \\
 &\quad + 77 + 73 + 80 + 76 + 74 + 79 + 72 + 81 + 78 \\
 &= 1898
 \end{aligned}$$

Therefore:

$$\bar{X} = \frac{1898}{25}$$
$$\boxed{\bar{X} = 75.92}$$

So, the **mean exam score is 75.92**.

Step 2: Calculate Standard Deviation

Since these 25 students are considered a **sample**, we use the **sample standard deviation**.

Formula:

$$s = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

We have:

$$\bar{X} = 75.92$$

The sum of squared deviations is:

$$\sum (X - \bar{X})^2 = 368.84$$

Therefore:

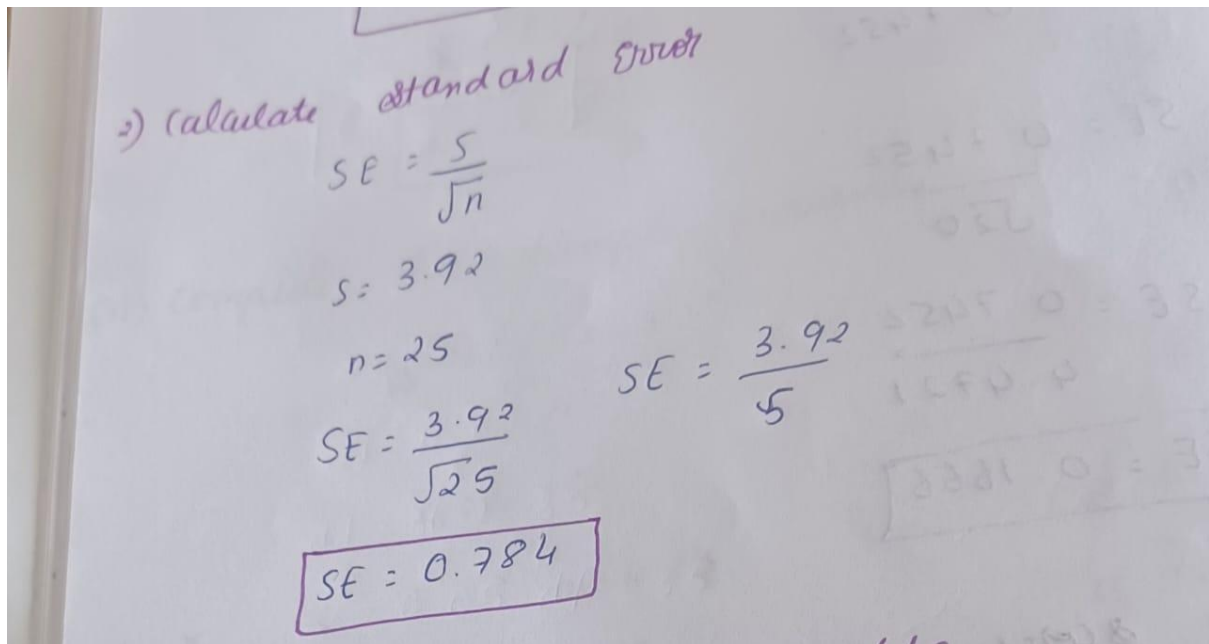
$$s = \sqrt{\frac{368.84}{25 - 1}}$$
$$s = \sqrt{\frac{368.84}{24}}$$
$$s = \sqrt{15.3683}$$
$$\boxed{s \approx 3.92}$$

Answer

$$\boxed{\text{Mean} = 75.92}$$

$$\boxed{\text{Standard Deviation} \approx 3.92}$$

2. Calculate Standard Error



A photograph of a piece of paper with handwritten calculations in purple ink. The text reads: '2) calculate standard error', followed by the formula $SE = \frac{s}{\sqrt{n}}$, then the values $s = 3.92$ and $n = 25$. Below these, the calculation $SE = \frac{3.92}{\sqrt{25}}$ is shown, and finally, the result $SE = 0.784$ is boxed.

The standard error of the mean is:

$$SE = \frac{s}{\sqrt{n}}$$

We know:

$$s = 3.92$$

and

$$n = 25$$

Therefore:

$$\begin{aligned} SE &= \frac{3.92}{\sqrt{25}} \\ SE &= \frac{3.92}{5} \\ SE &\approx 0.784 \end{aligned}$$

Answer

$$\boxed{\text{Standard Error} \approx 0.784}$$

3. Compare standard deviation and standard error.

Standard deviation and standard error are related but they measure different things.

Standard Deviation

The standard deviation measures the spread of individual student scores around the mean score.

In this problem:

$$SD \approx 3.92$$

This means that individual exam scores typically vary around the mean score of 75.92 by approximately 3.92 marks.

Standard Error

The standard error measures the variability or uncertainty of the sample mean as an estimate of the population mean.

In this problem:

$$SE \approx 0.784$$

The standard error is smaller than the standard deviation because:

$$SE = \frac{SD}{\sqrt{n}}$$

Here:

$$SE = \frac{3.92}{\sqrt{25}}$$
$$SE = \frac{3.92}{5} = 0.784$$

Problem 9: Manufacturing Weights

A sample of 50 packets has:

- Mean weight = 498 g
- Standard deviation = 12 g

Tasks:

1. Construct a 95% confidence interval for the population mean weight.

Problem 9: Manufacturing weights

$$\alpha = 1 - 0.95 = 0.05$$

$$df = n - 1 = 50 - 1 = 49$$

$$t_{0.025, 49} = 2.009$$

$$CI = \bar{x} \pm t \frac{s}{\sqrt{n}}$$

$$CI = 498 \pm 2.009 \left(\frac{12}{\sqrt{50}} \right)$$

$$\sqrt{50} = 7.071$$

$$\frac{12}{7.071} = 1.697$$

$$2.009 \times 1.697 = 3.409$$

$$CI = 498 \pm 3.409$$

$$\text{Lower Limit} = 498 - 3.409 = 494.591$$

$$\text{Upper Limit} = 498 + 3.409 = 501.409$$

$$95\% CI = (494.59, 501.41) g$$

$$494.59 g < \mu < 501.41 g$$

Since the sample size is **50** and the population standard deviation is not given, we use the **t-distribution**.

For a 95% confidence interval:

$$\alpha = 1 - 0.95 = 0.05$$

Degrees of freedom:

$$df = n - 1 = 50 - 1 = 49$$

For 95% confidence and 49 degrees of freedom:

$$t_{0.025,49} \approx 2.009$$

Formula

$$CI = \bar{X} \pm t \frac{s}{\sqrt{n}}$$

Substitute the values:

$$CI = 498 \pm 2.009 \left(\frac{12}{\sqrt{50}} \right)$$

First:

$$\begin{aligned} \sqrt{50} &\approx 7.071 \\ \frac{12}{7.071} &\approx 1.697 \end{aligned}$$

Now calculate the margin of error:

$$2.009 \times 1.697 \approx 3.409$$

Therefore:

$$CI = 498 \pm 3.409$$

Lower Limit

$$498 - 3.409 = 494.591$$

Upper Limit

$$498 + 3.409 = 501.409$$

Therefore:

$$\boxed{95\% CI = (494.59, 501.41) \text{ g}}$$

Final Answer

$$\boxed{494.59 \text{ g} < \mu < 501.41 \text{ g}}$$

So, the 95% confidence interval for the population mean weight is approximately **494.59 g to 501.41 g**.

2. Interpret the interval.

The 95% confidence interval is:

$$\boxed{494.59 \text{ g to } 501.41 \text{ g}}$$

This means that we are 95% confident that the true population mean weight of all packets lies between approximately 494.59 g and 501.41 g.

In simple words, although the sample mean is 498 g, the actual average weight of the entire population may be somewhat lower or higher. Based on this sample, the reasonable range for the population mean is approximately 494.59 g to 501.41 g.

3. Determine whether the target weight of 500 g is plausible.

The target weight is:

500 g

Our confidence interval is:

(494.59, 501.41)

Now check whether 500 g falls inside this interval:

$$494.59 < 500 < 501.41$$

Therefore:

500 g is inside the 95% confidence interval

Conclusion

Yes, the target weight of 500 g is plausible based on the sample data.

Because **500 g lies within the 95% confidence interval**, the sample does not provide sufficient evidence at the 5% significance level to conclude that the true population mean differs from 500 g.

Problem 10: Customer Satisfaction

Two branches report:

Branch Mean Rating 95% CI

A	4.3	(4.1, 4.5)
B	4.0	(3.8, 4.2)

Tasks:

1. Compare the Confidence Intervals

problem 10: Customer satisfaction

1) Compare the Confidence Intervals

Branch A	mean Rating	4.3	CI (4.1, 4.5)
Branch B	mean Rating	4.0	CI (3.8, 4.2)

Branch A

95% CI = (4.1, 4.5)

mean

4.3

Branch B

95% CI = (3.8, 4.2)

mean

4.0

overlapping A and B

$$-1 \leq z \leq 1$$

	A	B
Mean Rating	4.3	4.0
LL	4.1	3.8
UL	4.5	4.2

Branch A

$$95\% CI = (4.1, 4.5)$$

Mean:

4.3

Branch B

$$95\% CI = (3.8, 4.2)$$

Mean:

4.0

Comparison

	Branch A	Branch B
Mean Rating	4.3	4.0
Lower Limit	4.1	3.8
Upper Limit	4.5	4.2

The intervals are:

$$A = (4.1, 4.5)$$
$$B = (3.8, 4.2)$$

They **overlap** between 4.1 and 4.2.

$$\mu - 1\sigma \text{ to } \mu + 1\sigma$$

$$-1 \leq Z \leq 1$$

About 68.3% of values fall in this interval, from 1 below the mean to 1 above it

2. Can You Conclude Branch A Performs Better?

No, we cannot conclusively conclude that Branch A performs better than Branch B based only on these confidence intervals.

Although Branch A has a higher sample mean:

$$4.3 > 4.0$$

its 95% confidence interval is:

$$(4.1, 4.5)$$

while Branch B's interval is:

$$(3.8, 4.2)$$

The two confidence intervals **overlap from 4.1 to 4.2**.

Therefore, the observed difference in sample means does not by itself establish that Branch A's true population mean satisfaction is significantly higher.

3. Explain Your Reasoning

Branch A has a higher average rating of **4.3**, compared with **4.0** for Branch B. At first glance, this suggests that customers are more satisfied with Branch A.

However, confidence intervals show the **range of plausible values for the true population mean**. Branch A's true mean is estimated to lie between **4.1 and 4.5**, while Branch B's true mean is estimated to lie between **3.8 and 4.2**.

Because the intervals overlap between **4.1 and 4.2**, there is uncertainty about whether the difference between the two branches is statistically meaningful.

Important Point

Overlapping 95% confidence intervals do not automatically prove that the difference is statistically insignificant. To make a formal conclusion about whether Branch A is significantly better, we would need an appropriate **hypothesis test or confidence interval for the difference between the two means**.

Final Conclusion

Branch A has a higher observed mean satisfaction rating than Branch B (4.3 vs. 4.0). However, based only on the given 95% confidence intervals, we cannot conclusively state that Branch A performs better because the confidence intervals overlap. A formal statistical test comparing the two population means would be required for a definitive conclusion.

Problem 11: Average Battery Life

A battery manufacturer claims the average battery life is 10 hours.

A sample of 36 batteries gives:

- Mean = 9.5 hours
- Standard deviation = 1.8 hours

Tasks:

1. Formulate Null and Alternative Hypotheses

problem 11: Average Battery Life
2) conduct hypothesis test at $\alpha = 0.05$

$$n = 36$$

step I

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$t = \frac{9.5 - 10}{1.8/\sqrt{36}}$$

$$\sqrt{36} = 6$$

$$t = \frac{-0.5}{1.8/6}$$

$$t = \frac{-0.5}{0.3}$$

$$t = -1.667$$

$$z = \frac{\bar{x} - \mu}{\sigma}$$

$$z = \frac{(1.2) - (1.0)}{1} = 1.2$$

$$(z) = 88.5\%$$

step II:

$$\alpha = 0.05$$

$$df = n - 1$$

$$df = 36 - 1 = 35$$

$$\alpha = 0.05, df = 35$$

$$t_{\text{critical}} = 2.030$$

$$t < -2.030$$

$$t > 2.030$$

$$-2.030 < -1.667 < 2.030$$

step 3:

$$t = -1.667$$

$$-2.030, +2.030$$

do not reject H_0

The manufacturer **claims the average battery life is 10 hours**.

Since we want to test whether the actual average is **different from 10 hours**, this is a **two-tailed test**.

Null Hypothesis

$$H_0: \mu = 10$$

The population mean battery life is 10 hours.

Alternative Hypothesis

$$H_1: \mu \neq 10$$

The population mean battery life is not 10 hours.

Therefore:

$$\begin{array}{l} H_0: \mu = 10 \\ H_1: \mu \neq 10 \end{array}$$

2. Conduct Hypothesis Test at $\alpha = 0.05$

Since the sample size is:

$$n = 36$$

and the population standard deviation is not given, we can use a **one-sample t-test**.

Step 1: Calculate the Test Statistic

Formula:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

Substitute the values:

$$t = \frac{9.5 - 10}{1.8/\sqrt{36}}$$

Since:

$$\sqrt{36} = 6$$

we get:

$$t = \frac{-0.5}{1.8/6}$$
$$t = \frac{-0.5}{0.3}$$
$$t = -1.667$$

$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{(1.2) - (0)}{1} = 1.2$$

$$\Phi(z) \approx 88.5\%$$

Step 2: Find Critical Value

For a **two-tailed test**:

$$\alpha = 0.05$$

Degrees of freedom:

$$df = n - 1$$
$$df = 36 - 1 = 35$$

At:

$$\alpha = 0.05, df = 35$$

the critical t-values are approximately:

$$t_{critical} = \pm 2.030$$

So the rejection regions are:

$$t < -2.030$$

or

$$t > 2.030$$

Step 3: Compare Test Statistic and Critical Value

Calculated value:

$$t = -1.667$$

Critical values:

$$-2.030, + 2.030$$

Since:

$$-2.030 < -1.667 < 2.030$$

the test statistic **does not fall in the rejection region.**

Therefore:

Do not reject H_0

3. State Your Conclusion

At the **5% significance level**, the calculated test statistic is:

$$t = -1.667$$

and the critical values are:

$$\pm 2.030$$

Since the calculated t-value lies between the critical values, we **fail to reject the null hypothesis.**

Final Conclusion

There is **not enough statistical evidence at the 5% significance level to conclude that the population mean battery life is different from 10 hours.**

Although the sample mean is **9.5 hours**, which is lower than the claimed 10 hours, the difference is **not statistically significant** at the 5% level.

Therefore, based on this sample, the manufacturer's claim of an average battery life of **10 hours cannot be rejected**.

Problem 12: Marketing Campaign

A company believes a new campaign increases sales.

Sales before campaign:

50, 52, 49, 55, 53, 51, 54, 50

Sales after campaign:

58, 60, 57, 62, 59, 61, 63, 58

Tasks:

1. Conduct a Hypothesis Test

problem 12: Marketing Campaign

1. Step 1: Formulate Hypothesis

d : After - Before

$H_0: \mu_d = 0 \rightarrow$ Null

$H_1: \mu_d > 0 \rightarrow$ Alternative

The campaign increased average sales.

Step 2: Calculate Differences

$d = 8, 8, 8, 7, 6, 10, 9, 8$

$$\sum d = 64$$

Step 3: Mean Difference $\bar{d} = \frac{\sum d}{n}$

$$\bar{d} = 64/8$$

$$\bar{d} = 8$$

Step 4: Std. Dev Difference

$0, 0, 0, 1, 4, 4, 1, 0$

$$\sum (d - \bar{d})^2 = 10$$

$$S_d^2 = \frac{10}{8-1}$$

$$S_d^2 = \frac{10}{7} = 1.4286$$

$$S_d = \sqrt{1.4286}$$

$$S_d = 1.195$$

Step 5: Calculate Test Statistic

$$t = \frac{\bar{d}}{S_d / \sqrt{n}}$$

$$t = \frac{8}{1.195 / \sqrt{8}}$$

$$\sqrt{8} = 2.828$$

$$t = \frac{8}{1.195 / 2.828}$$

$$t = 8 / 0.423$$

$$t = 18.91$$

$$df = 8 - 1 = 7$$

$$df = 7$$

The company believes the campaign **increases sales**, so this is a **right-tailed paired t-test**.

Step 1: Formulate Hypotheses

Let:

$$d = \textit{After} - \textit{Before}$$

Null Hypothesis

$$H_0: \mu_d = 0$$

There is no increase in average sales after the campaign.

Alternative Hypothesis

$$H_1: \mu_d > 0$$

The campaign increases average sales.

Step 2: Calculate Differences

Before After Difference $d = \textit{After} - \textit{Before}$

50	58	8
52	60	8
49	57	8
55	62	7
53	59	6
51	61	10
54	63	9
50	58	8

Therefore:

$$d = 8, 8, 8, 7, 6, 10, 9, 8$$

Sum of differences:

$$\sum d = 64$$

Step 3: Mean Difference

$$\bar{d} = \frac{\sum d}{n}$$

$$\bar{d} = \frac{64}{8}$$

$$\boxed{\bar{d} = 8}$$

So, the average increase in sales is **8 units**.

Step 4: Standard Deviation of Differences

The deviations from the mean difference of 8 are:

$$0, 0, 0, -1, -2, 2, 1, 0$$

Squared deviations:

$$0, 0, 0, 1, 4, 4, 1, 0$$

Therefore:

$$\sum (d - \bar{d})^2 = 10$$

Sample variance:

$$s_d^2 = \frac{10}{8 - 1}$$

$$s_d^2 = \frac{10}{7} = 1.4286$$

Thus:

$$s_d = \sqrt{1.4286}$$

$$\boxed{s_d \approx 1.195}$$

Step 5: Calculate Test Statistic

For a paired t-test:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

Substitute:

$$t = \frac{8}{1.195/\sqrt{8}}$$
$$\sqrt{8} \approx 2.828$$
$$t = \frac{8}{1.195/2.828}$$
$$t = \frac{8}{0.423}$$
$$\boxed{t \approx 18.91}$$

The paired t-test uses $n - 1$ degrees of freedom.

$$df = 8 - 1 = 7$$

So:

$$\boxed{df = 7}$$

2. Calculate the p-value

Because the alternative hypothesis is:

$$H_1: \mu_d > 0$$

we use a **one-tailed p-value**.

With:

$$t \approx 18.91, df = 7$$

the p-value is extremely small:

$$\boxed{p \approx 0.0000001}$$

or approximately:

$$\boxed{p < 0.001}$$

3. Interpret the p-value

The p-value is approximately:

$$p < 0.001$$

A p-value tells us how much evidence the data provide against the null hypothesis. A very small p-value indicates strong evidence against the null hypothesis.

Here:

$$p < 0.001$$

which is much smaller than the significance level:

$$\alpha = 0.05$$

Therefore, there is **very strong statistical evidence that the marketing campaign increased sales**.

4. Make a Decision at $\alpha = 0.05$

Compare:

$$p < 0.001$$

$$\alpha = 0.05$$

$$p < \alpha$$

$$\text{Reject } H_0$$

Final Conclusion

At the 5% significance level, we reject the null hypothesis. There is sufficient statistical evidence to conclude that the new marketing campaign **significantly increases sales**.

The sample shows an average increase of:

$$8 \text{ sales units}$$

Problem 13: Call Centre Response Time

A sample of 100 calls has:

- Mean response time = 4.8 minutes
- Standard deviation = 1.5 minutes

Tasks:

1. Estimate the Population Mean

problem 13: Call Centre Response Time:

mean response time = 4.8 mins

standard deviation = 1.5 mins

2) $n = 100$

$$z_{0.025} = 1.96$$

$$CI = \bar{x} \pm z \frac{s}{\sqrt{n}}$$

$$-1 \leq z \leq 1$$

Step I: calculate std. E

$$SE = s/\sqrt{n}$$

$$SE = \frac{1.5}{\sqrt{100}}$$

$$SE = 1.5/10$$

$$SE = 0.15$$

Step II: calculate margin of error

$$ME = 1.96 \times 0.15$$

$$ME = 0.294$$

Step III: Calculate confidence interval

$$CI = 4.8 \pm 0.294$$

$$4.8 - 0.294 = 4.506$$

$$4.8 + 0.294 = 5.094$$

$$95\% CI = (4.506, 5.094) \text{ mins}$$

$$95\% CI = (4.51, 5.09) \text{ mins}$$

There is a 95% probability that the true population mean b/w 4.51 and 5.09.

The **sample mean** is used as the point estimate of the population mean.

$$\boxed{\mu \approx \bar{X}}$$

Given:

$$\bar{X} = 4.8$$

Therefore:

$$\boxed{\text{Estimated population mean} = 4.8 \text{ minutes}}$$

So, the estimated average response time for the entire population of calls is **4.8 minutes**.

2. Construct a 95% Confidence Interval

Since the sample size is large:

$$n = 100$$

we can use the **z-value** for a 95% confidence interval:

$$z_{0.025} = 1.96$$

The formula is:

$$CI = \bar{X} \pm z \frac{s}{\sqrt{n}}$$

$$\mu - 1\sigma \quad \mu + 1\sigma$$

$$-1 \leq Z \leq 1$$

About 68.3% of values fall in this interval, from 1 below the mean to 1 above it

Step 1: Calculate Standard Error

$$\begin{aligned} SE &= \frac{s}{\sqrt{n}} \\ SE &= \frac{1.5}{\sqrt{100}} \\ SE &= \frac{1.5}{10} \end{aligned}$$

$$SE = 0.15$$

Step 2: Calculate Margin of Error

$$ME = 1.96 \times 0.15$$

$$ME = 0.294$$

Step 3: Calculate Confidence Interval

$$CI = 4.8 \pm 0.294$$

Lower limit:

$$4.8 - 0.294 = 4.506$$

Upper limit:

$$4.8 + 0.294 = 5.094$$

Therefore:

$$95\% CI = (4.506, 5.094) \text{ minutes}$$

Approximately:

$$95\% CI = (4.51, 5.09) \text{ minutes}$$

3. Explain the Frequentist Interpretation

The sample gives an estimated population mean response time of:

$$4.8 \text{ minutes}$$

and the 95% confidence interval is:

$$(4.51, 5.09) \text{ minutes}$$

Frequentist Interpretation

In the **Frequentist approach**, the population mean is considered a fixed but unknown value, while the confidence interval changes from sample to sample.

The correct interpretation is:

If we repeatedly took random samples of 100 calls and constructed a 95% confidence interval using the same method each time, approximately 95% of those intervals would contain the true population mean response time.

For this particular sample, we obtain an interval from approximately **4.51 minutes to 5.09 minutes**.

Important Point

It is **not technically correct** to say:

"There is a 95% probability that the true population mean is between 4.51 and 5.09 minutes."

In Frequentist statistics, the population mean is fixed; the **interval-producing procedure** has the 95% coverage property.

Problem 14: Defective Products

A sample of 500 products contains 22 defective items.

Tasks:

1. Estimate the Defect Rate

problem 14: Defective products

1. Estimate the Defect Rate

$$\hat{p} = \frac{x}{n}$$

$$\hat{p} = 22/500$$

$$\hat{p} = 0.044$$

$$0.044 \times 100 = 4.4\%$$

$$\hat{p} = 0.044$$

Estimated defect Rate = 4.4%

2) Construct a 95% Confidence interval

$$CI = \hat{p} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$z = 1.96$$

$$37\% = \frac{37}{100} = 0.37$$

Step I: Calculate Standard Error

$$SE = \sqrt{\frac{0.044(1-0.044)}{500}}$$

$$SE = \sqrt{\frac{0.044(0.956)}{500}}$$

$$SE = \sqrt{\frac{0.042064}{500}}$$

$$SE = \sqrt{0.000084128}$$

$$SE = 0.00917$$

Step II: Calculate margin of Error

$$ME = 1.96 \times 0.00917$$

$$ME = 0.01797$$

step 3: calculate confidence interval

$$CI = 0.044 \pm 0.01797$$

$$0.044 - 0.01797 = 0.02603$$

$$0.044 + 0.01797 = 0.06197$$

$$95\% CI = (0.0260, 0.0620)$$

$$0.0260 \times 100 = 2.60\%$$

$$0.0620 \times 100 = 6.20\%$$

$$95\% CI = (2.60\%, 6.20\%)$$

The **sample proportion** is used as the point estimate of the population defect rate.

The formula is:

$$\hat{p} = \frac{x}{n}$$

Substitute the values:

$$\hat{p} = \frac{22}{500}$$

$$\hat{p} = 0.044$$

Convert to percentage:

$$0.044 \times 100 = 4.4\%$$

Therefore:

$$\hat{p} = 0.044$$

or

$$\boxed{\text{Estimated defect rate} = 4.4\%}$$

So, approximately **4.4% of the products are estimated to be defective.**

2. Construct a 95% Confidence Interval

For a population proportion, the 95% confidence interval is:

$$CI = \hat{p} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

For 95% confidence:

$$z = 1.96$$

$$37\% = \frac{37}{100} = 0.37$$

Percent of the whole

%

Percent of the whole

Step 1: Calculate Standard Error

$$SE = \sqrt{\frac{0.044(1 - 0.044)}{500}}$$

$$SE = \sqrt{\frac{0.044(0.956)}{500}}$$

$$SE = \sqrt{\frac{0.042064}{500}}$$

$$SE = \sqrt{0.000084128}$$

$$\boxed{SE \approx 0.00917}$$

Step 2: Calculate Margin of Error

$$ME = 1.96 \times 0.00917$$

$$\boxed{ME \approx 0.01797}$$

Step 3: Calculate Confidence Interval

$$CI = 0.044 \pm 0.01797$$

Lower Limit

$$0.044 - 0.01797 = 0.02603$$

Upper Limit

$$0.044 + 0.01797 = 0.06197$$

Therefore:

$$95\% CI = (0.0260, 0.0620)$$

Convert to percentages:

$$0.0260 \times 100 = 2.60\%$$

$$0.0620 \times 100 = 6.20\%$$

Final Answer

$$95\% CI = (2.60, \% 6.20\%)$$

So, the estimated population defect rate is **4.4%**, with a 95% confidence interval of approximately **2.60% to 6.20%**.

3. Explain the Frequentist Meaning of the Interval

The estimated defect rate from the sample is:

$$4.4\%$$

and the 95% confidence interval is:

$$(2.60, \%6.20\%)$$

The **Frequentist interpretation** is:

If we repeatedly took random samples of 500 products from the same population and constructed a 95% confidence interval using the same method each time, approximately 95% of those intervals would contain the true population defect rate.

For this particular sample, the interval obtained is **2.60% to 6.20%**.

Problem 15: Email Spam Detection

Historical information:

- Probability an email is spam = 0.25
- Probability spam email contains keyword "Offer" = 0.80
- Probability non-spam email contains keyword "Offer" = 0.10

Tasks:

1. Compute the probability that an email is spam given it contains "Offer."
2. Apply Bayes' theorem.
3. Interpret the result.

1. Compute the Probability That an Email Is Spam Given It Contains "Offer"

We need to calculate:

$$P(S | O)$$

where:

- S = email is spam
- O = email contains "Offer"

$$P(\text{Spam}) = 0.25$$

$$P(\text{Not Spam}) = 1 - 0.25 = 0.75$$

$$P(\text{Offer} | \text{Spam}) = 0.80$$

$$P(\text{Offer} | \text{Not Spam}) = 0.10$$

We need to find:

$$P(\text{Spam} | \text{Offer})$$

2. Apply Bayes' Theorem

Bayes' Theorem

$$P(S | O) = \frac{P(O | S)P(S)}{P(O)}$$

First, we need to calculate $P(O)$, the probability that an email contains "Offer".

Using the **law of total probability**:

$$P(O) = P(O | S)P(S) + P(O | S^c)P(S^c)$$

Substitute the values:

$$\begin{aligned}P(O) &= (0.80)(0.25) + (0.10)(0.75) \\P(O) &= 0.20 + 0.075 \\P(O) &= 0.275\end{aligned}$$

Now apply Bayes' theorem:

$$\begin{aligned}P(S | O) &= \frac{(0.80)(0.25)}{0.275} \\P(S | O) &= \frac{0.20}{0.275} \\P(S | O) &= 0.72727\end{aligned}$$

Convert to percentage:

$$0.72727 \times 100 = 72.727\%$$

Therefore:

$$\boxed{P(S | O) \approx 72.73\%}$$

3. Interpret the Result

The probability that an email is spam **given that it contains the keyword "Offer"** is approximately:

$$\boxed{72.73\%}$$

This means that when an email contains the word **"Offer"**, there is approximately a **72.73% chance that the email is spam**, based on the given historical probabilities.

The result also shows why **Bayes' theorem is useful in spam detection**. The keyword "Offer" alone does not automatically mean that an email is spam. Instead, Bayes' theorem combines:

- The initial probability of spam (**25%**).
- How frequently spam emails contain "Offer" (**80%**).
- How frequently non-spam emails contain "Offer" (**10%**).

After considering all this information, the probability increases from **25% before seeing the keyword** to approximately **72.73% after observing "Offer"**.

Problem 16: Disease Diagnosis

Given:

- Disease prevalence = 2%
- Test sensitivity = 95%
- Test specificity = 90%

Tasks:

1. **Compute the posterior probability of disease after a positive test.**

problem 16: Disease Diagnosis

$$p(D) = 0.02$$

$$p(+|D) = 0.95$$

$$p(-|D') = 0.90$$

$$p(D') = 1 - 0.02 = 0.98$$

posterior probability of disease after a positive test.

$$p(D|+)$$

$$p(D|+) = \frac{p(+|D)p(D)}{p(+|D)p(D) + (p(+|D')p(D'))}$$

$$p(D|+) = \frac{(0.95)(0.02)}{(0.95)(0.02) + (0.10)(0.98)}$$

$$= \frac{0.019}{0.019 + 0.098}$$

$$= \frac{0.019}{0.117}$$

$$p(D|+) = 0.1624$$

$$p(D|+) = 16.24\%$$

We use Bayes' theorem:

$$P(\text{Disease} | \text{Positive}) = [P(\text{Positive} | \text{Disease}) \times P(\text{Disease})] / P(\text{Positive})$$

First calculate $P(\text{Positive})$:

$$P(\text{Positive}) = [P(\text{Positive} | \text{Disease}) \times P(\text{Disease})] + [P(\text{Positive} | \text{No Disease}) \times P(\text{No Disease})]$$

$$= (0.95 \times 0.02) + (0.10 \times 0.98)$$

$$= 0.019 + 0.098$$

$$= 0.117$$

Now apply Bayes' theorem:

$$P(\text{Disease} | \text{Positive}) = (0.95 \times 0.02) / 0.117$$

$$= 0.019 / 0.117$$

$$= 0.1624$$

Convert into percentage:

$$0.1624 \times 100 = 16.24\%$$

Therefore,

Posterior probability = 16.24%

2. Explain Why the Result Differs From 95%

The **95% sensitivity** and **16.24% posterior probability** are measuring two different things.

Sensitivity of 95% means:

If a person actually has the disease, there is a 95% chance that the test will correctly give a positive result.

Posterior probability of 16.24% means:

If a person receives a positive test result, there is only a 16.24% probability that the person actually has the disease.

The result is much lower than 95% mainly because the disease is **rare**. The disease prevalence is only **2%**.

Also, the test has a **10% false-positive rate** because its specificity is 90%.

So, even though the test is good at detecting disease, many healthy people can also receive positive results.

Problem 17: Website Conversion Rate

A company tests two landing pages.

Group Visitors Purchases

A 5000 250

B 5200 312

Tasks:

1. Calculate conversion rates.

2. Formulate hypotheses.
3. Conduct a two-proportion z-test.
4. Determine whether page B performs better.

problem 17: website conversion rate

(i) calculate conversion rate

$$\text{Conversion Rate} = \frac{\text{purchases}}{\text{visitors}} \times 100$$
$$PA = \frac{250}{5000} \times 100$$
$$PA = 5\%$$

$$P_B = \frac{312}{5200} \times 100$$

$$P_B = 6\%$$

vii) conduct two - proportion z - test

$$P_A = \frac{250}{5000} = 0.05$$

$$P_B = \frac{312}{5200} = 0.06$$

$$P = \frac{250 + 312}{5000 + 5200}$$

$$P = \frac{562}{10200} = 0.0551$$

$$q = 1 - P$$

$$q = 1 - 0.0551 = 0.9449$$

$$SE = \sqrt{Pq \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}$$

$$SE = \sqrt{(0.0551)(0.9449) \left(\frac{1}{5000} + \frac{1}{5200} \right)}$$

$$SE = 0.00452$$

$$Z = \frac{P_B - P_A}{SE}$$

$$Z = \frac{0.06 - 0.05}{0.00452}$$

$$Z = 2.21$$

1. Calculate Conversion Rates

Conversion rate is calculated as:

$$\text{Conversion Rate} = (\text{Number of Purchases} / \text{Number of Visitors}) \times 100$$

Group A

$$\text{Conversion Rate A} = (250 / 5000) \times 100$$

$$= 0.05 \times 100$$

$$= \mathbf{5\%}$$

Group B

$$\text{Conversion Rate B} = (312 / 5200) \times 100$$

$$= 0.06 \times 100$$

$$= \mathbf{6\%}$$

Therefore:

Group A conversion rate = 5%

Group B conversion rate = 6%

Page B has a conversion rate that is **1 percentage point higher** than Page A.

2. Formulate Hypotheses

We want to determine whether **Page B performs better than Page A**.

Let:

p_A = population conversion rate for Page A

p_B = population conversion rate for Page B

Null Hypothesis

$$H_0: p_A = p_B$$

There is no significant difference between the conversion rates of Page A and Page B.

Alternative Hypothesis

$$H_1: p_B > p_A$$

Page B has a higher conversion rate than Page A.

This is a **right-tailed two-proportion z-test**.

3. Conduct a Two-Proportion Z-Test

Step 1: Calculate Sample Proportions

$$\hat{p}_A = 250 / 5000$$

$$\hat{p}_A = \mathbf{0.05}$$

$$\hat{p}_B = 312 / 5200$$

$$\hat{p}_B = \mathbf{0.06}$$

Step 2: Calculate the Pooled Proportion

Under the null hypothesis, we assume both pages have the same population conversion rate.

Formula:

$$\hat{p} = (x_A + x_B) / (n_A + n_B)$$

Substitute:

$$\hat{p} = (250 + 312) / (5000 + 5200)$$

$$\hat{p} = 562 / 10200$$

$$\hat{p} = \mathbf{0.055098}$$

So:

$$\hat{p} \approx 0.0551$$

Step 3: Calculate the Standard Error

Formula:

$$SE = \sqrt{\hat{p}(1 - \hat{p})(1/n_A + 1/n_B)}$$

Substitute:

$$SE = \sqrt{[0.055098(1 - 0.055098)(1/5000 + 1/5200)]}$$

$$SE = \sqrt{[0.055098 \times 0.944902 \times (0.0002 + 0.0001923)]}$$

$$SE \approx \sqrt{0.00002043}$$

$$SE \approx \mathbf{0.00452}$$

Step 4: Calculate the Z-Statistic

Formula:

$$z = (\hat{p}_B - \hat{p}_A) / SE$$

Substitute:

$$z = (0.06 - 0.05) / 0.00452$$

$$z = 0.01 / 0.00452$$

$$z \approx \mathbf{2.21}$$

Therefore:

$$\mathbf{z \approx 2.21}$$

Step 5: Find the Critical Value

For a right-tailed test at:

$$\alpha = 0.05$$

The critical z-value is:

$$z_{\text{critical}} = 1.645$$

Now compare:

$$\text{Calculated } z = 2.21$$

$$\text{Critical } z = 1.645$$

Since:

$$2.21 > 1.645$$

we **reject H_0** .

The one-tailed p-value is approximately:

$$p \approx 0.0136$$

Since:

$$0.0136 < 0.05$$

we also reject H_0 .

4. Determine Whether Page B Performs Better

Page A has a conversion rate of:

$$5\%$$

Page B has a conversion rate of:

$$6\%$$

Although Page B has a higher observed conversion rate, we need to determine whether this difference is statistically significant.

The two-proportion z-test gives:

$$z \approx 2.21$$

and

$$p\text{-value} \approx 0.0136$$

Since:

$$p\text{-value} < \alpha$$

$$0.0136 < 0.05$$

we reject the null hypothesis.

Final Conclusion

There is **sufficient statistical evidence at the 5% significance level to conclude that Page B performs better than Page A.**

Page B has a conversion rate of **6%**, compared with **5%** for Page A. The difference is statistically significant according to the two-proportion z-test.

Problem 18: Mobile App Feature Test

Feature A:

- 400 users
- Average engagement = 18 min
- SD = 5 min

Feature B:

- 420 users
- Average engagement = 20 min
- SD = 6 min

Tasks:

Problem 13: Mobile App Feature Test

(i) Calculate the test statistic

$$Z = \frac{\bar{y}_B - \bar{y}_A}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

$$Z = \frac{20 - 18}{\sqrt{\frac{5^2}{400} + \frac{6^2}{420}}}$$

$$Z = \frac{2}{\sqrt{\frac{25}{400} + \frac{36}{420}}}$$

$$Z = \frac{2}{\sqrt{0.0625 + 0.0857}}$$

$$Z = \frac{2}{\sqrt{0.1482}}$$

$$Z = \frac{2}{0.385}$$

$$Z = 5.19$$

(ii) Calculate the p-value

$$P = P(Z > 5.19)$$

$$P = 1.0 \times 10^{-7}$$

$$P = 0.0000001$$

$$P < 0.05$$

1. Test whether Feature B increases engagement.
2. Calculate test statistic and p-value.
3. Interpret results.

Problem 19: Recommendation System

A new recommendation algorithm increases click-through rate from 4.8% to 5.3%.

Sample size:

- Old system: 100,000 users
- New system: 100,000 users

Tasks:

problem 19: Recommendation System

1. Hypothesized

$P_1 = 0.048 \rightarrow$ old system CTR

$P_2 = 0.053 \rightarrow$ new system CTR

$$H_0: P_1 = P_2$$

$$H_1: P_1 \neq P_2$$

2. calculate the z-statistic

$$p = \frac{P_1 n_1 + P_2 n_2}{n_1 + n_2}$$

$$p = \frac{(0.048)(100000) + (0.053)(100000)}{200000}$$

$$p = 0.0505$$

$$q = 1 - p = 0.9495$$

$$SE = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$SE = \sqrt{(0.0505)(0.9495) \left(\frac{1}{100000} + \frac{1}{100000} \right)}$$

$$SE = 0.000979$$

$$Z = \frac{P_2 - P_1}{SE}$$

$$Z = \frac{0.053 - 0.048}{0.000979}$$

$$Z = 5.11$$

calculate p value

$$p\text{-value} = 2P(Z > 5.11)$$

$$p = 0.00000033$$

$$p < 0.05$$

$$5.3\% - 4.8\% = 0.5\%$$

$$\frac{5.3 - 4.8}{4.8} \times 100 = 10.42\%$$

$$Z = 5.11$$

$$P = 0.000000033$$

1. Test Whether Feature B Increases Engagement

Let:

μ_A = population mean engagement for Feature A

μ_B = population mean engagement for Feature B

Since we want to know whether B is **greater than A**, this is a **right-tailed two-sample test**.

Null Hypothesis

$H_0: \mu_B = \mu_A$

There is no increase in engagement with Feature B.

Alternative Hypothesis

$H_1: \mu_B > \mu_A$

Feature B increases average engagement.

2. Calculate Test Statistic and p-value

Since the two groups are independent and their standard deviations are given, we use a **two-sample z-test**.

The test statistic is:

$$z = (\bar{x}_B - \bar{x}_A) / \sqrt{[(s_A^2/n_A) + (s_B^2/n_B)]}$$

Substitute the values:

$$z = (20 - 18) / \sqrt{[(5^2/400) + (6^2/420)]}$$

$$z = 2 / \sqrt{[(25/400) + (36/420)]}$$

$$z = 2 / \sqrt{[0.0625 + 0.085714]}$$

$$z = 2 / \sqrt{0.148214}$$

$$z = 2 / 0.38499$$

$$z \approx \mathbf{5.195}$$

Therefore:

Test statistic, $z \approx 5.20$

Calculate p-value

Since this is a **right-tailed test**:

$$\text{P-value} = P(Z > 5.20)$$

The p-value is approximately:

$$\mathbf{p\text{-value} \approx 0.0000001}$$

Therefore:

$$\mathbf{p\text{-value} < 0.001}$$

3. Interpret the Results

The calculated test statistic is:

$$\mathbf{z \approx 5.20}$$

The p-value is:

$$\mathbf{p < 0.001}$$

At the 5% significance level:

$$\alpha = 0.05$$

Since:

$$\text{p-value} < \alpha$$

p-value < 0.05

we reject the null hypothesis.

Conclusion

There is **strong statistical evidence** that **Feature B** increases average user engagement compared with **Feature A**.

Problem 20: Salary Prediction Model

Two machine learning models produce Mean Absolute Errors (MAE):

Model MAE

Model A 4200

Model B 4000

Across 50 test folds, the difference in MAE has:

- Mean difference = 200
- Standard deviation = 350

Tasks:

problem 20: salary prediction model

1. Hypotheses

$$d = MAE_A - MAE_B$$
$$H_0: \mu_d = 0$$
$$H_1: \mu_d > 0$$
$$t = \frac{\bar{d} - 0}{sd / \sqrt{n}}$$
$$t = \frac{200}{350 / \sqrt{50}}$$
$$\sqrt{50} = 7.071$$
$$t = \frac{200}{350 / 7.071}$$
$$t = \frac{200}{49.497}$$

$$t = 4.04$$

$$df = n - 1$$

$$df = 50 - 1 = 49$$

$$t = 4.04, df = 49$$

$$p = 0.0001$$

$$p < 0.05$$

$$MAE_A = 4200$$

$$MAE_B = 4000$$

$$4200 - 4000 = 200$$

$$t(49) = 4.04, p = 0.0001$$

1. Perform a Paired t-Test

First, define the difference as:

Difference = MAE of Model A - MAE of Model B

Given:

Mean difference = 200

This is positive because:

$$4200 - 4000 = 200$$

A positive difference means that **Model B has a lower MAE**, which is better.

Step 1: Formulate Hypotheses

Null hypothesis:

$$H_0: \mu_d = 0$$

There is no significant difference between the MAE of Model A and Model B.

Alternative hypothesis:

$$H_1: \mu_d > 0$$

Model B has significantly lower MAE than Model A.

This is a **one-tailed paired t-test**.

Step 2: Calculate the Test Statistic

Formula:

$$t = \text{Mean Difference} / (\text{Standard Deviation of Differences} / \sqrt{n})$$

Substitute the values:

$$t = 200 / (350 / \sqrt{50})$$

$$\sqrt{50} = 7.071$$

$$t = 200 / (350 / 7.071)$$

$$t = 200 / 49.497$$

$$t \approx \mathbf{4.04}$$

Therefore:

$$\mathbf{\text{Test statistic} = 4.04}$$

Step 3: Degrees of Freedom

$$df = n - 1$$

$$df = 50 - 1$$

$$df = \mathbf{49}$$

Step 4: Calculate the p-value

For:

$$t = 4.04$$

$$df = 49$$

and a one-tailed test:

$$\mathbf{p\text{-value} \approx 0.000094}$$

Therefore:

$$\mathbf{p\text{-value} < 0.001}$$

2. Determine Whether Model B Significantly Outperforms Model A

The significance level is:

$$\alpha = 0.05$$

The calculated p-value is:

$$p \approx 0.000094$$

Compare:

$p\text{-value} < \alpha$

$0.000094 < 0.05$

Therefore, we **reject the null hypothesis**.

Conclusion

There is sufficient statistical evidence to conclude that **Model B significantly outperforms Model A** in terms of MAE.

The average MAE difference is:

$$4200 - 4000 = \mathbf{200}$$

Since lower MAE means better prediction accuracy, Model B is better by an average of **200 MAE units**.

3. Explain the Result in Terms of Statistical Significance

The p-value is approximately:

0.000094

This is much smaller than the significance level of 0.05.

This means that the observed improvement in Model B is **very unlikely to have occurred simply due to random variation between the test folds**, assuming there is actually no difference between the models.

Therefore, the difference in MAE is **statistically significant**.

In simple words:

- Model A MAE = **4200**
- Model B MAE = **4000**
- Difference = **200**
- Lower MAE is better.
- The p-value is **less than 0.05**.
- Therefore, **Model B is statistically significantly better than Model A**.